Practical 5: Xen Virtualization with XenCenter

Subject: Cloud Computing | **Course:** M.Sc. IT Part I, Semester I

Aim: Implement Xen virtualization and manage it with XenCenter.

Requirements

- VMware Workstation (to run XenServer as a VM)
- XenServer ISO (Citrix XenServer / Citrix Hypervisor)
- Citrix XenCenter installer, which is a Windows application
- Windows XP SP3 ISO, shared from a folder on your PC
- Virtualization (VT-x/AMD-V) enabled in BIOS, and at least 8 GB RAM

Part A: Install XenServer in VMware Workstation

1. Open VMware Workstation and choose **File → New Virtual Machine → Custom (advanced)**.
2. Choose **Installer disc image file (ISO)** and select the XenServer ISO.
3. Set the guest OS to **Linux** (a Linux version such as "Other Linux 64-bit" or "CentOS 64-bit").
4. Name the VM `XenServer`. Give it **2 or more processors**, **4 GB or more RAM** and a **60 GB or larger** disk.
5. Set the network to **Bridged** or **NAT**.
6. Under the processor settings, tick **Virtualize Intel VT-x/EPT or AMD-V/RVI**.
7. Power on the VM and run the installer:
   - Select the keyboard layout.
   - Accept the EULA.
   - Select the disk to install on.
   - Choose **Local media** as the installation source.
   - Set the **root password** (the journal uses `root123`).
   - Set a static or DHCP IP and a hostname.
   - Select the time zone and the NTP/manual time setting.

8. Confirm the installation and let it finish. Reboot when prompted.
9. The blue XenServer console screen then shows the **IP address** (for example `192.168.124.137`). Ping it from your PC's command prompt to confirm it is reachable.

Part B: Install XenCenter and add the server

1. Install **Citrix XenCenter** on your Windows PC if it isn't already installed.
2. Open XenCenter and click **Add a Server**.
3. Enter the XenServer IP, the username `root` and the password you set, then click **Add**.
4. The host now appears in the left pane.

Part C: Add storage for the ISO

1. Click **New Storage**.
2. Choose **Windows File Sharing (CIFS)** and click **Next**.
3. Untick **Auto-generate** the description, give it a name, and click **Next**.
4. Enter the share path of the folder containing the Windows XP ISO, for example `\\<PC-IP>\isos`. Enter your local PC credentials and click **Finish**.

Part D: Create and start the Windows XP VM

1. Click **New VM**, select **Windows XP SP3**, and click **Next**.
2. Name the VM, choose the **ISO** from the storage you added, and click **Next**.
3. Accept the default CPU, memory and disk settings, clicking **Next** each time.
4. Untick **Start the new VM automatically** and click **Create Now**.
5. Right-click the XP VM and choose **Start**, then open the **Console** tab.
6. Complete the Windows XP setup. When the XP welcome screen appears, the practical is successful.

Output (screenshots to paste)

1. The XenServer installer and the console screen showing the IP
2. The ping reply
3. The server added in XenCenter

4. The CIFS storage repository
5. The VM creation wizard
6. The Windows XP welcome screen